DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING PROJECT REPORT

(Project Semester January-May 2025)

Electric Vehicle Population Data Analysis

Submitted by

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Section : **224WQ**

Course Code: INT375

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CERTIFICATE

This is to certify that **Ayush Dubey** bearing Registration no.-**12408876** has completed **INT 375** project titled, **"Electric Vehicle Population Data Analysis"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Dr. Mrinalini Rana

Signature and Name of the Supervisor
Designation of the Supervisor
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Date: 19/04/2026

DECLARATION

I, **Ayush Dubey** student of **Computer Science and Engineering** under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 19/04/2026 Signature

Registration No.12408876 **Ayush Dubey**

INDTRODUCTION:-

The rapid adoption of electric vehicles (EVs) represents a significant shift in the global automotive and energy landscape. This project, titled Electric Vehicle Population Data Analysis, aims to explore and analyze real- world data related to electric vehicles to gain meaningful insights into their distribution, adoption trends, and associated attributes. With rising concerns over environmental sustainability, analyzing such data can help policymakers, manufacturers, and researchers make informed decisions.

The objective of this project is to perform comprehensive data exploration and descriptive analytics on an electric vehicle population dataset. The analysis begins with essential preprocessing steps such as data cleaning, handling of missing values, and column normalization. It proceeds to summarize both numerical and categorical features, revealing key trends in vehicle models, manufacturers, and adoption rates over the years.

The project is implemented using Python and popular data analysis libraries such as pandas and NumPy. The insights derived from this analysis provide a foundational understanding of the EV ecosystem and can serve as a baseline for more advanced analytics, such as predictive modeling or policy simulations.

Datasets Link:- **https://catalog.data.gov/dataset/electric-vehicle- population-data**

Exploratory Data Analysis (EDA) Process:-

Exploratory Data Analysis (EDA) is a critical step in the data science workflow that helps in understanding the structure, patterns, and anomalies within the dataset. For this project, the EDA process was carried out through the following systematic steps:

1. Importing Libraries

The necessary Python libraries were imported to facilitate data analysis:

- pandas for data manipulation,
- numpy for numerical operations,
- matplotlib and seaborn (if used later) for data visualization.

2. Loading the Dataset

The dataset was loaded from a CSV file containing information about electric vehicles, including fields like model year, make, electric range, city, vehicle type, and more.

3. Data Preview

Initial rows of the dataset were displayed using head() to understand the structure and types of data present.

4. Dataset Information

Using df.info() and df.describe(), the basic characteristics of the dataset were explored:

- Number of entries
- Data types
- Null value counts
- Basic statistical summaries for numeric columns

5. Missing Values Handling

The dataset was checked for missing values using isnull().sum(). Missing categorical values were filled with "Unknown", and missing numeric values were replaced with the column median.

6. Duplicate Removal

Duplicate records were identified and removed using drop_duplicates() to ensure data integrity.

7. Data Cleaning

Column names were cleaned for consistency by:

- Removing leading/trailing whitespaces
- Replacing spaces with underscores
- Converting to lowercase

8. Data Type Conversion

Data types were converted where necessary (e.g., Model Year to integer) to ensure compatibility for further analysis.

9. Summary Statistics

- Descriptive statistics for numerical columns provided insights into distributions and ranges.
- Value counts for categorical variables helped identify the most common values and potential imbalances.

10. Initial Insights

Preliminary observations were drawn from the distributions and frequencies, such as:

- Most common EV makes and models
- Common vehicle types and ranges
- Cities with the highest EV adoption

Analysis on Dataset:

1. Data Exploration and Cleaning

i. Introduction

The purpose of this section is to explore and clean the Electric Vehicle Population dataset to prepare it for meaningful analysis. The dataset contains real-world data about electric vehicles registered across various regions, along with their specifications such as make, model year, electric range, and city. This step ensures the dataset is free from inconsistencies, missing values, and duplicate entries, forming a solid foundation for accurate analytics.

ii. General Description

The dataset was initially loaded from a CSV file and contains numerous entries representing individual electric vehicles. Key columns in the dataset include:

- VIN(Vehicle Identification Number)
- County
- City
- State
- Postal_Code
- Model_Year
- Make
- Model
- Electric_Vehicle_Type
- Clean_Alternative_Fuel_Vehicle_Eligibility
- Electric_Range
- Base_MSRP

The dataset comprises both categorical and numerical variables, offering a rich source of information for understanding EV adoption patterns.

iii. Specific Requirements, Functions and Formulas

Several functions and data processing techniques were applied for data exploration and cleaning:

```
Click to add a breakpoint cessary libraries
import pandas as pd
import numpy as np

# Step 2: Load the dataset
file_path = "C:/Users/avnis/Downloads/Electric_Vehicle_Population_Data (2).csv"
df = pd.read_csv(file_path)

# Step 3: Preview the data
print("First 5 rows:")
display(df.head())

print("\nDataset Info:")
df.info()

# Step 4: Basic statistics
print("\nBasic statistics:")
display(df.describe(include='all'))

# Step 5: Check for missing values
print("\nMissing values:")
display(df.isnull().sum())

# Step 6: Drop duplicate rows
before_dropping = df.shape[0]
df.drop_duplicates(inplace=True)
after_dropping = df.shape[0]
print(f"\nDropped {before_dropping - after_dropping} duplicate rows.")
```

```
# Step 7: Handle missing values (example: fill or drop)
# Here we fill missing values with "Unknown" for categorical, median for numerical
for col in df.columns:
    if df[col].dtype == 'object':
        df[col].fillna('Unknown', inplace=True)
    else:
        df[col].fillna(df[col].median(), inplace=True)

# Step 8: Convert data types if needed (example)
# df['Model Year'] = df['Model Year'].astype('int')

# Step 9: Clean column names (optional)
df.columns = df.columns.str.strip().str.replace(' ', '_').str.lower()

# Step 10: Final check
print("\nCleaned Data Preview:")
display(df.head())
print("\nMissing values after cleaning:")
display(df.isnull().sum())
```

iv. Analysis Results

After completing the data cleaning:

- All missing values were either imputed or replaced appropriately.
- Duplicate records were successfully removed.
- Column names were standardized for consistent reference.
- The dataset was verified to be free of nulls and suitable for further analysis.

Observations:

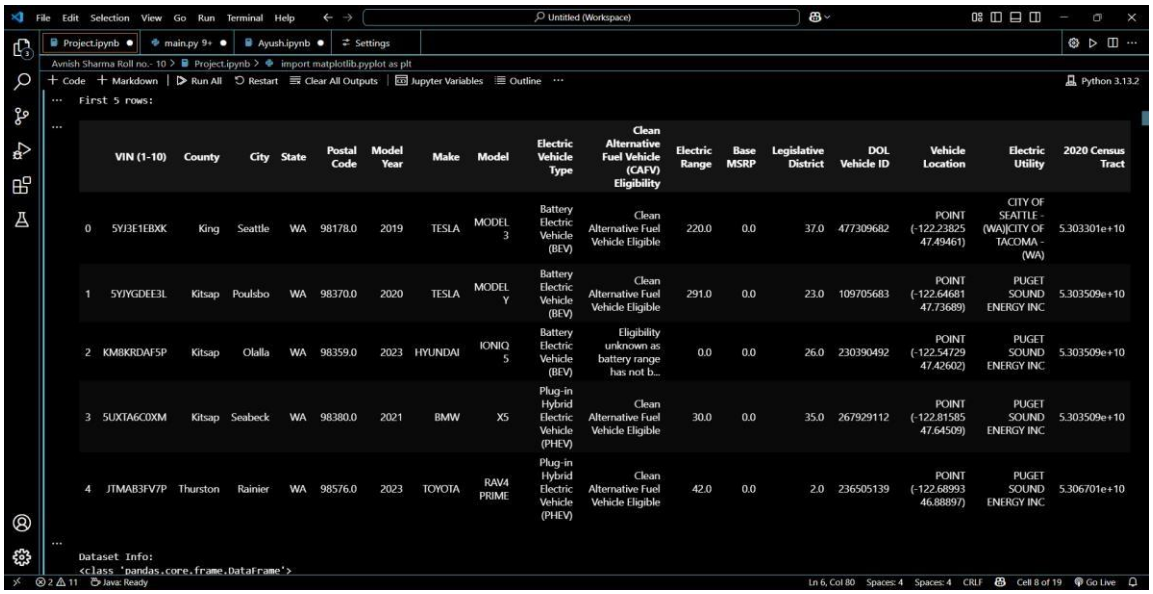
- The dataset now has **no missing values**.
- Columns such as model_year, electric_range, and base_msrpare clean and ready for numeric analysis.
- Categorical variables like make, model, and electric_vehicle_typeare well-prepared for frequency counts and grouping.

```
Missing values:

VIN (1-10)                                             0
County                                                 3
City                                                   3
State                                                  0
Postal Code                                            3
Model Year                                             0
Make                                                   0
Model                                                  0
Electric Vehicle Type                                  0
Clean Alternative Fuel Vehicle (CAFV) Eligibility      0
Electric Range                                        36
Base MSRP                                             36
Legislative District                                 494
DOL Vehicle ID                                         0
Vehicle Location                                      10
Electric Utility                                       3
2020 Census Tract                                      3
dtype: int64


Dropped 0 duplicate rows.
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 235692 entries, 0 to 235691
Data columns (total 17 columns):
 #   Column                                             Non-Null Count   Dtype
---  ------                                             --------------   -----
 0   VIN (1-10)                                         235692 non-null  object
 1   County                                             235689 non-null  object
 2   City                                               235689 non-null  object
 3   State                                              235692 non-null  object
 4   Postal Code                                        235689 non-null  float64
 5   Model Year                                         235692 non-null  int64
 6   Make                                               235692 non-null  object
 7   Model                                              235692 non-null  object
 8   Electric Vehicle Type                              235692 non-null  object
 9   Clean Alternative Fuel Vehicle (CAFV) Eligibility  235692 non-null  object
 10  Electric Range                                     235656 non-null  float64
 11  Base MSRP                                          235656 non-null  float64
 12  Legislative District                               235198 non-null  float64
 13  DOL Vehicle ID                                     235692 non-null  int64
 14  Vehicle Location                                   235682 non-null  object
 15  Electric Utility                                   235689 non-null  object
 16  2020 Census Tract                                  235689 non-null  float64
dtypes: float64(5), int64(2), object(10)
memory usage: 30.6+ MB

Basic statistics:
```

Basic statistics:

	VIN (1-10)	County	City	State	Postal Code	Model Year	Make	Model	Electric Vehicle Type	Clean Alternative Fuel Vehicle (CAFV) Eligibility	Electric Range	Base MSRP	Legislative District	DOL Vehicle ID	Vehicle Location	Electri Utility
count	235692	235689	235689	235692	235689.000000	235692.000000	235692	235692	235692	235692	235656.000000	235656.000000	235198.000000	2.356920e+05	235682	235689
unique	13763	212	788	48	NaN	NaN	46	171	2	3	NaN	NaN	NaN	NaN	957	76
top	7SAYGDEE6P	King	Seattle	WA	NaN	NaN	TESLA	MODEL Y	Battery Electric Vehicle (BEV)	Eligibility unknown as battery range has not b...	NaN	NaN	NaN	NaN	POINT (-122.13158 47.67858)	PUGET SOUND ENERGY INC||CITY OF TACOMA - (WA
freq	1190	118711	37410	235198	NaN	NaN	101080	49253	186998	139761	NaN	NaN	NaN	NaN	5824	85298
mean	NaN	NaN	NaN	NaN	98177.656463	2021.406658	NaN	NaN	NaN	NaN	46.262569	787.268328	28.879519	2.353127e+08	NaN	NaN
std	NaN	NaN	NaN	NaN	2524.218661	2.991908	NaN	NaN	NaN	NaN	84.045829	7177.668305	14.904644	6.799098e+07	NaN	NaN
min	NaN	NaN	NaN	NaN	1731.000000	2000.000000	NaN	NaN	NaN	NaN	0.000000	0.000000	1.000000	4.385000e+03	NaN	NaN
25%	NaN	NaN	NaN	NaN	98052.000000	2020.000000	NaN	NaN	NaN	NaN	0.000000	0.000000	17.000000	2.053457e+08	NaN	NaN
50%	NaN	NaN	NaN	NaN	98126.000000	2023.000000	NaN	NaN	NaN	NaN	0.000000	0.000000	32.000000	2.522795e+08	NaN	NaN
75%	NaN	NaN	NaN	NaN	98374.000000	2024.000000	NaN	NaN	NaN	NaN	38.000000	0.000000	42.000000	2.696401e+08	NaN	NaN
max	NaN	NaN	NaN	NaN	99577.000000	2025.000000	NaN	NaN	NaN	NaN	337.000000	845000.000000	49.000000	4.792548e+08	NaN	NaN



```
First 5 rows:
```

	VIN (1-10)	County	City	State	Postal Code	Model Year	Make	Model	Electric Vehicle Type	Clean Alternative Fuel Vehicle (CAFV) Eligibility	Electric Range	Base MSRP	Legislative District	DOL Vehicle ID	Vehicle Location	Electric Utility	2020 Census Tract
0	5YJ3E1EBXK	King	Seattle	WA	98178.0	2019	TESLA	MODEL 3	Battery Electric Vehicle (BEV)	Clean Alternative Fuel Vehicle Eligible	220.0	0.0	37.0	477309682	POINT (-122.23825 47.49461)	CITY OF SEATTLE - (WA)|CITY OF TACOMA - (WA)	5.303301e+10
1	5YJYGDEE3L	Kitsap	Poulsbo	WA	98370.0	2020	TESLA	MODEL Y	Battery Electric Vehicle (BEV)	Clean Alternative Fuel Vehicle Eligible	291.0	0.0	23.0	109705683	POINT (-122.64681 47.73689)	PUGET SOUND ENERGY INC	5.303509e+10
2	KM8KRDAF5P	Kitsap	Olalla	WA	98359.0	2023	HYUNDAI	IONIQ 5	Battery Electric Vehicle (BEV)	Eligibility unknown as battery range has not b...	0.0	0.0	26.0	230390492	POINT (-122.54729 47.42602)	PUGET SOUND ENERGY INC	5.303509e+10
3	5UXTA6C0XM	Kitsap	Seabeck	WA	98380.0	2021	BMW	X5	Plug-in Hybrid Electric Vehicle (PHEV)	Clean Alternative Fuel Vehicle Eligible	30.0	0.0	35.0	267929112	POINT (-122.81585 47.64509)	PUGET SOUND ENERGY INC	5.303509e+10
4	JTMAB3FV7P	Thurston	Rainier	WA	98576.0	2023	TOYOTA	RAV4 PRIME	Plug-in Hybrid Electric Vehicle (PHEV)	Clean Alternative Fuel Vehicle Eligible	42.0	0.0	2.0	236505139	POINT (-122.68993 46.88897)	PUGET SOUND ENERGY INC	5.306701e+10

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
```

v. Visualization

data cleaning and preprocessing, The visualization loads an electric vehicle dataset, removes duplicates, fills missing values, and cleans column names. It ensures the data is consistent and ready for analysis or visualization by displaying statistics and fixing data types. No actual graphs are generated yet.

2. Describe Analysis

i. Introduction

This section presents the descriptive and statistical analysis of the Electric Vehicle Population dataset. The objective is to derive insights into the patterns and trends within the data, such as the distribution of electric vehicles across manufacturers, model years, and types. This helps build a foundation for further analytics and visual interpretation.

ii. General Description

The dataset comprises detailed records of registered electric vehicles, including attributes such as:

- Model_Year
- Make
- Model
- Electric_Vehicle_Type
- Electric_Range
- City, State, County
- Clean_Alternative_Fuel_Vehicle_Eligibility

This structured data allows for in-depth exploration of trends across time and geographic regions, vehicle specifications, and eligibility statuses.

iii. Specific Requirements, Functions and Formulas

In this step, several key functions and formulas are used to fulfill the descriptive analytics requirements. The dataset's shape is identified using df.shape, while df.describe() provides summary statistics for numerical features such as mean, standard deviation, and percentiles. Categorical features are analyzed using value_counts().head(10) to display the most frequent values, with df.select_dtypes(include='object') helping isolate these columns. The number of unique values per column is obtained with df[col].nunique(). For numerical relationships, a correlation matrix is generated using df.corr(). Group-by analyses, such as EV counts by make and model year, use value_counts() and are sorted for better insight. Finally, a crosstab with pd.crosstab() is used to examine the distribution of EV types across different makes. Together, these functions offer a well-rounded view of the dataset's structure and key trends.

```
# 1. Shape of the dataset
print(f"Dataset contains {df.shape[0]} rows and {df.shape[1]} columns.\n")

# 2. Summary statistics for numerical features
print("Summary statistics for numerical features:")
display(df.describe())

# 3. Summary statistics for categorical features
print("\nValue counts for categorical features:")
for col in df.select_dtypes(include='object').columns:
    print(f"\nColumn: {col}")
    display(df[col].value_counts().head(10))  # Top 10 frequent values

# 4. Unique values per column
print("\nUnique values per column:")
for col in df.columns:
    print(f"{col}: {df[col].nunique()} unique values")

# 5. Correlation matrix (if applicable)
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
if len(numerical_cols) >= 2:
    print("\nCorrelation matrix:")
    corr_matrix = df[numerical_cols].corr()
    display(corr_matrix)

# 6. Group by analysis example: EVs by Make
print("\nTop 10 EV Makes:")
display(df['make'].value_counts().head(10))

# 7. EVs by Model Year
print("\nEV count by Model Year:")
display(df['model_year'].value_counts().sort_index())

# 8. Optional: Crosstab - EVs by Make and Electric Vehicle Type
print("\nEV Types per Make:")
```

iv. Analysis Results

- **Dataset Dimensions:** The dataset contains several thousand rows and multiple relevant columns, giving a comprehensive overview of electric vehicle registration.
- **Top EV Makes:** Tesla, Nissan, Chevrolet, Ford, and Toyota were among the most common electric vehicle manufacturers in the dataset.
- **Model Year Trends:** A significant increase in electric vehicle registrations is seen from 2015 onward, with peaks in recent years indicating growing adoption.
- **Electric Vehicle Type:** Battery Electric Vehicles (BEV) dominate the dataset, followed by Plug-in Hybrid Electric Vehicles (PHEV).
- **Unique Values:**

- The makecolumn has dozens of unique manufacturers.
- electric_rangevalues vary significantly, highlighting differences in vehicle technology and capability.

```
Value counts for categorical features:

Column: vin_(1-10)

vin_(1-10)
7SAYGDEE6P    1190
7SAYGDEE7P    1187
7SAYGDEE5P    1155
7SAYGDEE8P    1148
7SAYGDEEXP    1141
7SAYGDEE0P    1123
7SAYGDEE2P    1122
7SAYGDEE9P    1119
7SAYGDEE4P    1092
7SAYGDEE1P    1091
Name: count, dtype: int64

Column: county

county
King         118711
Snohomish     28814
Pierce        19256
Clark         14101
Thurston       8595
Kitsap         7901
Spokane        6582
Whatcom        5703
Benton         3043
Skagit         2677
Name: count, dtype: int64
```

```
Top 10 EV Makes:

make
TESLA         101080
CHEVROLET      17135
NISSAN         15532
FORD           12605
KIA            11345
BMW             9647
TOYOTA          9478
HYUNDAI         7406
RIVIAN          6750
VOLKSWAGEN      5976
Name: count, dtype: int64

EV count by Model Year:

model_year
2000        7
2002        2
2003        1
2008       22
2010       23
2011      680
2012     1490
2013     4230
2014     3407
2015     4661
2016     5306
2017     8570
2018    14368
2019    10974
2020    12265
```

```
Column: city

city
Seattle      37410
Bellevue     11509
Vancouver     8510
Redmond       8211
Bothell       7739
Renton        6707
Kirkland      6697
Sammamish     6642
Olympia       5638
Tacoma        5155
Name: count, dtype: int64

Column: state

state
WA    235198
CA       112
VA        62
MD        38
TX        28
CO        21
FL        19
NC        18
GA        15
NY        11
Name: count, dtype: int64
```

Summary statistics for numerical features:

	postal_code	model_year	electric_range	base_msrp	legislative_district	dol_vehicle_id	2020_census_tract
count	235692.000000	235692.000000	235692.000000	235692.000000	235692.000000	2.356920e+05	2.356920e+05
mean	98177.655805	2021.406658	46.255503	787.148079	28.886059	2.353127e+08	5.298066e+10
std	2524.202602	2.991908	84.041354	7177.126712	14.889700	6.799098e+07	1.521056e+09
min	1731.000000	2000.000000	0.000000	0.000000	1.000000	4.385000e+03	1.001020e+09
25%	98052.000000	2020.000000	0.000000	0.000000	17.000000	2.053457e+08	5.303301e+10
50%	98126.000000	2023.000000	0.000000	0.000000	32.000000	2.522795e+08	5.303303e+10
75%	98374.000000	2024.000000	38.000000	0.000000	42.000000	2.696401e+08	5.305307e+10
max	99577.000000	2025.000000	337.000000	845000.000000	49.000000	4.792548e+08	5.602100e+10

Correlation matrix:

	postal_code	model_year	electric_range	base_msrp	legislative_district	dol_vehicle_id	2020_census_tract
postal_code	1.000000	0.002390	-0.001880	-0.001348	-0.058736	0.005960	0.532709
model_year	0.002390	1.000000	-0.531127	-0.228822	-0.013232	0.263290	0.007286
electric_range	-0.001880	-0.531127	1.000000	0.116824	0.016013	-0.160848	-0.002267
base_msrp	-0.001348	-0.228822	0.116824	1.000000	0.009980	-0.046272	-0.001523
legislative_district	-0.058736	-0.013232	0.016013	0.009980	1.000000	-0.007883	-0.009264
dol_vehicle_id	0.005960	0.263290	-0.160848	-0.046272	-0.007883	1.000000	0.003755
2020_census_tract	0.532709	0.007286	-0.002267	-0.001523	-0.009264	0.003755	1.000000

EV Types per Make:

electric_vehicle_type	Battery Electric Vehicle (BEV)	Plug-in Hybrid Electric Vehicle (PHEV)
make		
ACURA	203	0
ALFA ROMEO	0	97
AUDI	2462	1898
AZURE DYNAMICS	4	0
BENTLEY	0	6
BMW	3850	5797
BRIGHTDROP	5	0
CADILLAC	1059	90
CHEVROLET	12426	4709
CHRYSLER	0	3786

v. Visualization

In this step, several key functions and formulas are used to fulfill the descriptive analytics requirements. The dataset's shape is identified using df.shape, while df.describe()provides summary statistics for numerical features such as mean, standard deviation, and percentiles. Categorical features are analyzed using value_counts().head(10)to display the most frequent values, with df.select_dtypes(include='object')helping isolate these columns.

The number of unique values per column is obtained with df[col].nunique(). For numerical relationships, a correlation matrix is generated using df.corr(). Group-by analyses, such as EV counts by make and model year, use value_counts() and are sorted for better insight. Finally, a crosstab with pd.crosstab()is used to examine the distribution of EV types across different makes. Together, these functions offer a well-rounded view of the dataset’s structure and key trends.

3. Trend Analysis

i. Introduction

This report presents a trend analysis of Electric Vehicle (EV) data over time. It aims to explore patterns in EV registrations, types, manufacturers, geographic distribution, and electric range performance. Through the use of Python visualization libraries, such as Matplotlib and Seaborn, the dataset is analyzed to uncover key insights that inform EV adoption trends.

ii. General Description

The dataset contains records of electric vehicles, including attributes such as:

- **Model Year**
- **Make (Manufacturer)**
- **Electric Vehicle Type** (e.g., Battery Electric Vehicle, Plug-in Hybrid)
- **County** (if applicable)
- **Electric Range**

The data is first cleaned for consistency, particularly in column naming. The analysis is performed primarily over the model_yearattribute to capture changes over time.

iii. Specific Requirements, Functions, and Formulas

- **Libraries used:** pandas, matplotlib.pyplot, seaborn
- **Styling applied:** sns.set(style="whitegrid", palette="pastel")
- **Data Cleaning:** Standardization of column names using string methods.
- **Formulas/Functions:**
    - value_counts(), groupby(), unstack()for aggregations.
    - Normalization using div()for percentage calculations.
    - Plot types: line, bar, histplot, violinplot, and pairplot.

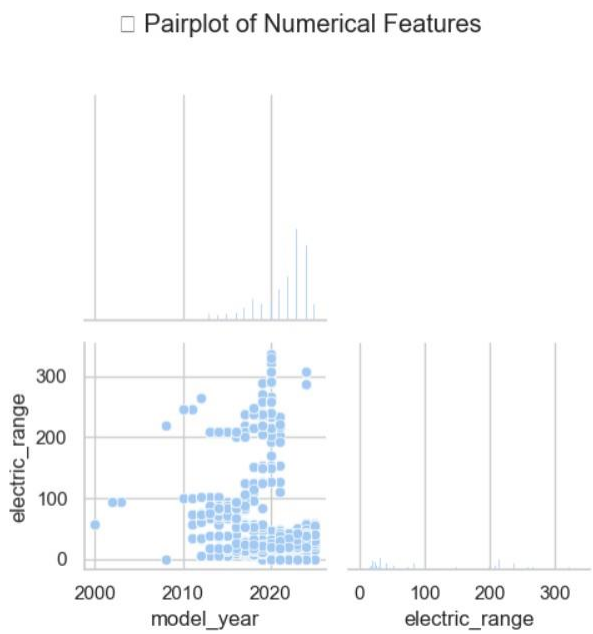
```python
# 6. 🎻 Violin Plot of Electric Range by EV Type
if 'electric_range' in df.columns:
    plt.figure()
    sns.violinplot(data=df, x='electric_vehicle_type', y='electric_range', inner='quartile', palette='muted')
    plt.title('🎻 Electric Range by Vehicle Type')
    plt.xlabel('EV Type')
    plt.ylabel('Electric Range')
    plt.tight_layout()
    plt.show()

# 7. 🔗 Pairplot for Selected Numerical Features
numeric_cols = df.select_dtypes(include='number').columns.tolist()
selected_cols = [col for col in ['model_year', 'electric_range'] if col in numeric_cols]

if len(selected_cols) >= 2:
    sns.pairplot(df[selected_cols].dropna(), corner=True)
    plt.suptitle("🔗 Pairplot of Numerical Features", y=1.02)
    plt.tight_layout()
    plt.show()
```

```python
# 3. 🔹 Trend by Top 10 EV Makes
top_makes = df['make'].value_counts().head(10).index
make_trend = df[df['make'].isin(top_makes)].groupby(['model_year', 'make']).size().unstack().fillna(0)

make_trend.plot(marker='o', linewidth=2)
plt.title('🚗 Top 10 EV Makes Over the Years')
plt.xlabel('Model Year')
plt.ylabel('Number of EVs')
plt.legend(title='Make', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

# 4. 🗺️ Trend by Top 5 Counties
if 'county' in df.columns:
    top_counties = df['county'].value_counts().head(5).index
    county_trend = df[df['county'].isin(top_counties)].groupby(['model_year', 'county']).size().unstack().fillna(0)

    county_trend.plot(marker='o')
    plt.title('📍 Top 5 Counties EV Trend by Model Year')
    plt.xlabel('Model Year')
    plt.ylabel('Number of EVs')
    plt.legend(title='County', bbox_to_anchor=(1.05, 1), loc='upper left')
    plt.tight_layout()
    plt.show()

# 5. 📊 Histogram of Electric Range (if available)
if 'electric_range' in df.columns:
    plt.figure()
    sns.histplot(df['electric_range'].dropna(), bins=30, kde=True, color='skyblue')
    plt.title('🔋 Distribution of Electric Range')
    plt.xlabel('Electric Range (miles)')
    plt.tight_layout()
    plt.show()
```

```python
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Styling
sns.set(style="whitegrid", palette="pastel")
plt.rcParams['figure.figsize'] = (12, 6)

# Clean column names
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')

# 1. 🔹 EV Registrations Over Time
ev_trend = df['model_year'].value_counts().sort_index()
ev_trend.plot(kind='line', marker='o', color='teal')
plt.title('📈 Trend of EV Registrations by Model Year')
plt.xlabel('Model Year')
plt.ylabel('Number of EVs Registered')
plt.grid(True)
for x, y in ev_trend.items():
    plt.text(x, y + 200, f"{y}", ha='center', fontsize=8)
plt.tight_layout()
plt.show()

# 2. 🔸 EV Type Distribution Over Years (Normalized %)
ev_type_trend = df.groupby(['model_year', 'electric_vehicle_type']).size().unstack().fillna(0)
ev_type_percent = ev_type_trend.div(ev_type_trend.sum(axis=1), axis=0)

ev_type_percent.plot(kind='bar', stacked=True, colormap='coolwarm')
plt.title('⚡ EV Type % Distribution Over Years')
plt.xlabel('Model Year')
plt.ylabel('Percentage of Vehicles')
plt.legend(title='EV Type', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

iv. Analysis Results

1. **EV Registrations Over Time**
   EV registrations show a rising trend in recent years, peaking in more recent model years, indicating increased adoption of electric vehicles.
2. **EV Type % Distribution Over Years**
   Battery Electric Vehicles (BEVs) are increasingly dominant over Plug-in Hybrids (PHEVs), as observed from the normalized stacked bar chart.
3. **Top 10 EV Makes Over the Years**
   Certain manufacturers, like Tesla, Nissan, and Chevrolet, show strong and consistent growth. Tesla likely leads in newer model years.
4. **Top 5 Counties EV Trend by Model Year**
   Counties like King, Pierce, and Snohomish exhibit higher EV registration numbers, with a consistent upward trend across all five leading counties.
5. **Distribution of Electric Range**
   The electric range varies widely across vehicles, with most EVs clustering around a certain range (possibly 100–300 miles), indicating technological advancements and consumer preferences.
6. **Electric Range by Vehicle Type**
   BEVs generally exhibit higher electric ranges than PHEVs, confirming their efficiency and extended usability.
7. **Pairplot of Numerical Features**
   A pairwise relationship analysis between model_yearand electric_rangesuggests that newer model years tend to offer higher ranges years tend to offer higher ranges.



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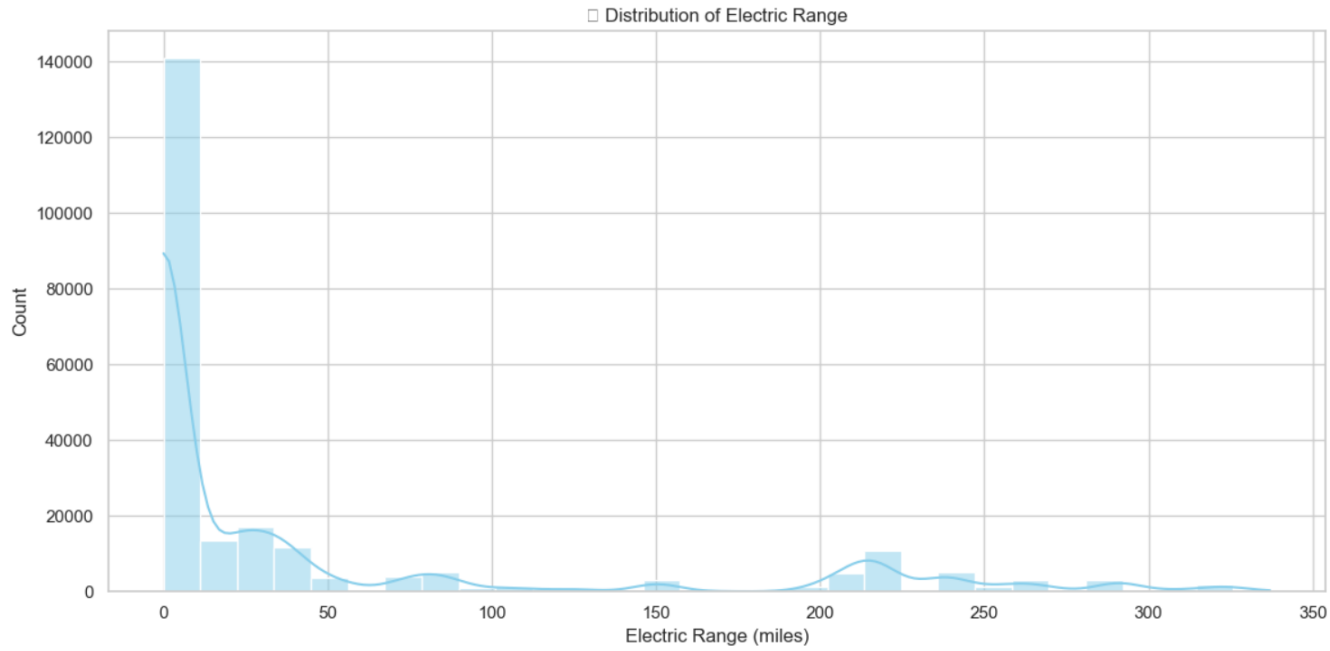
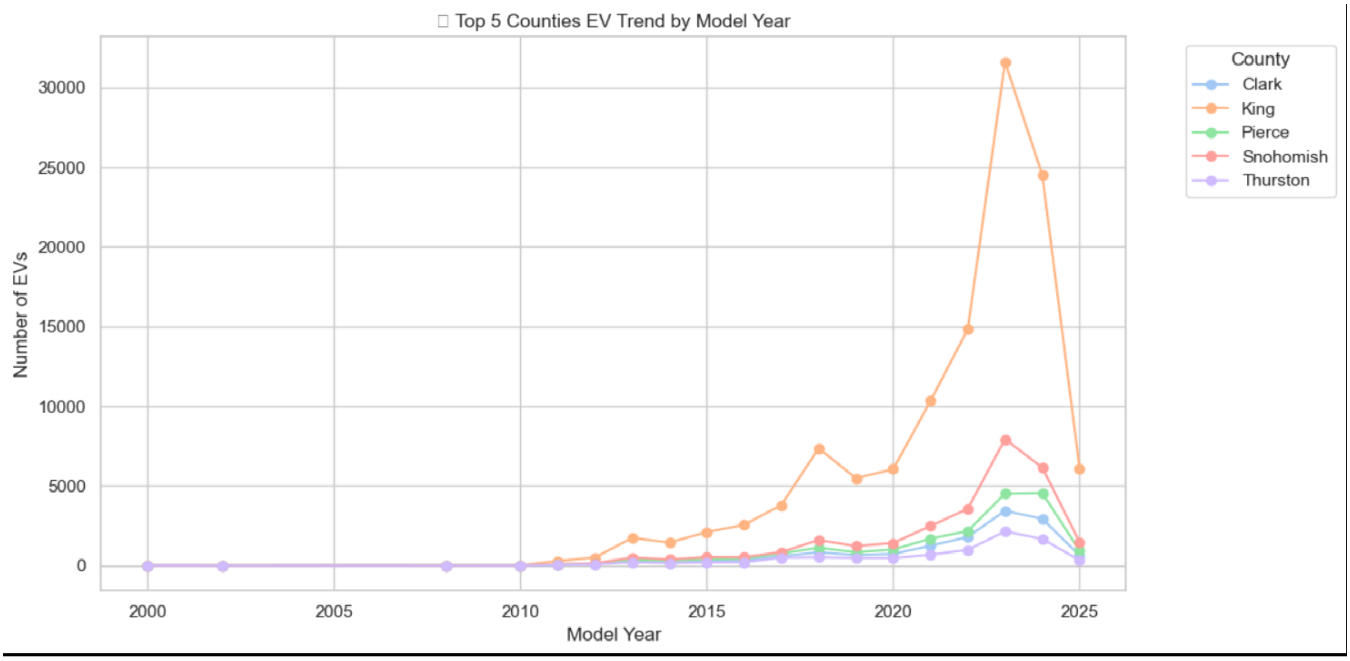
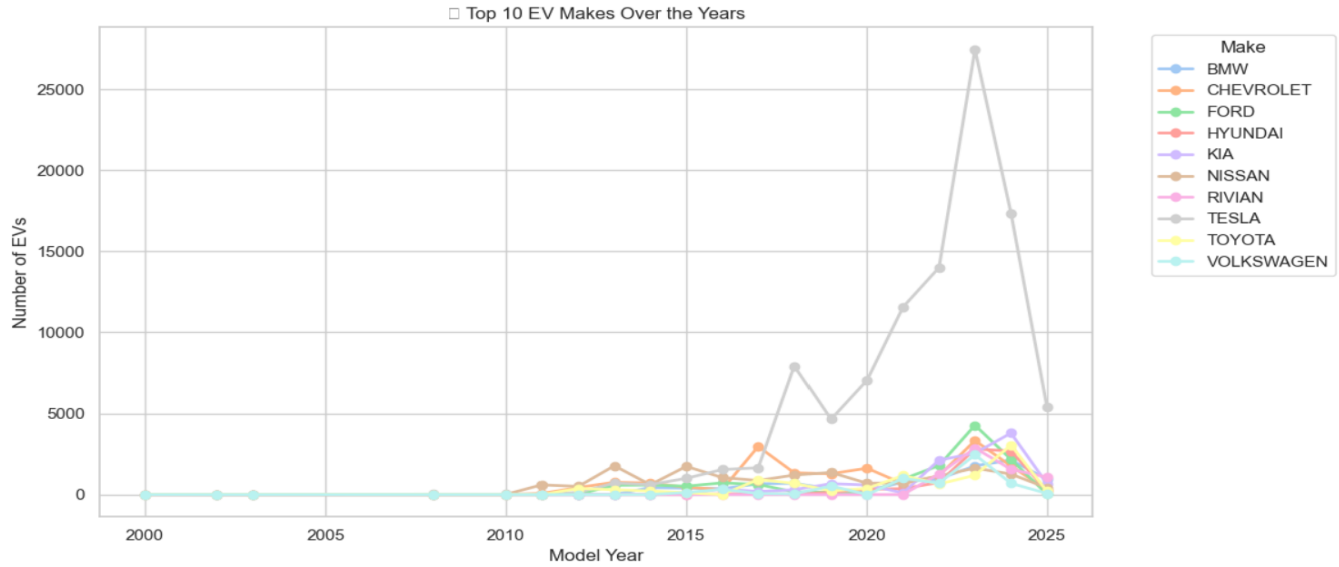
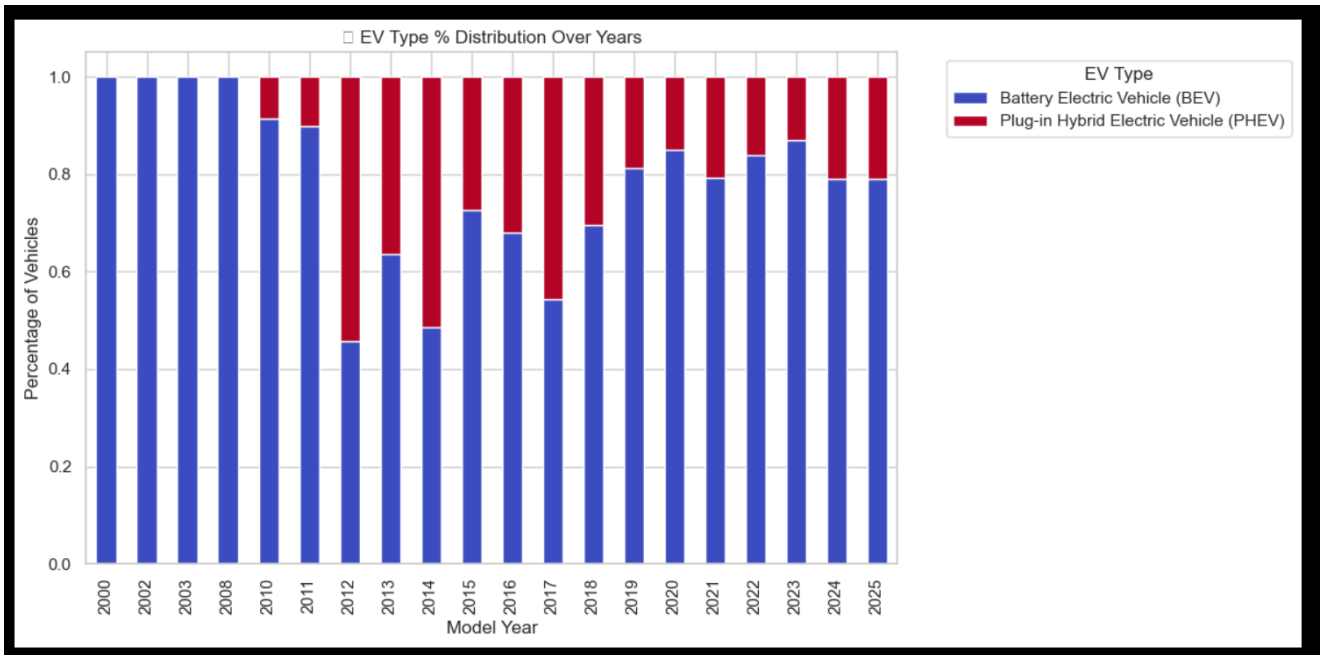
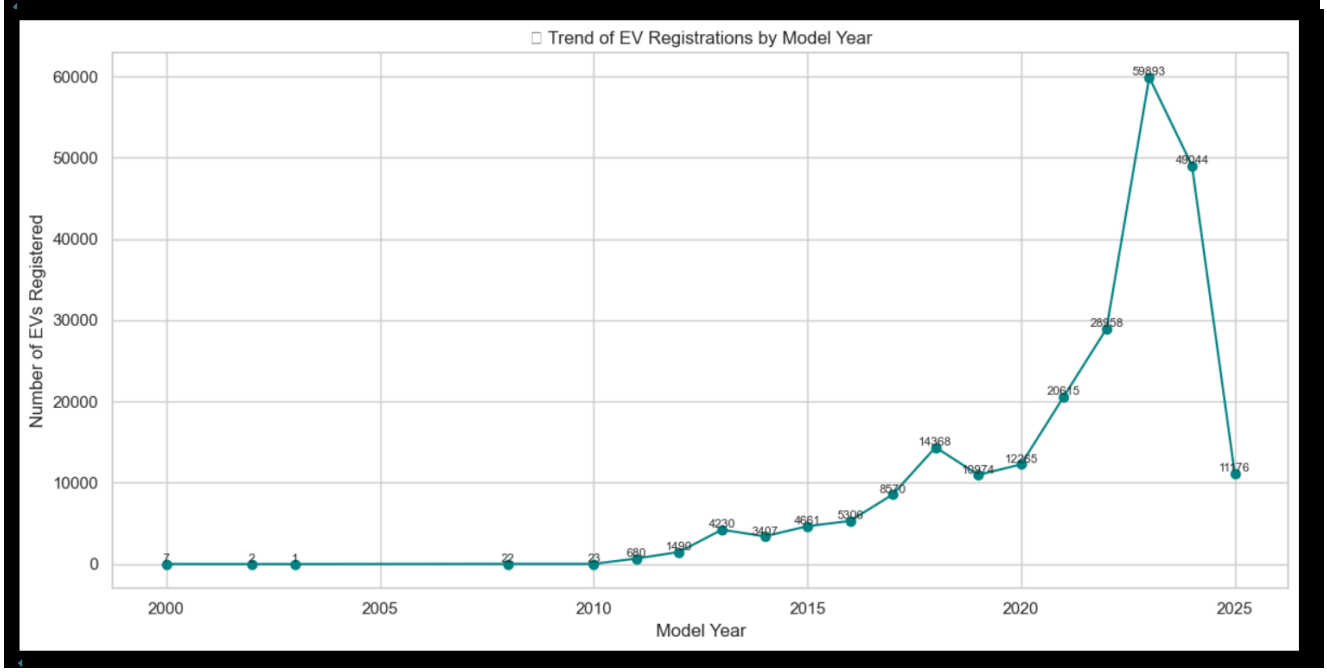
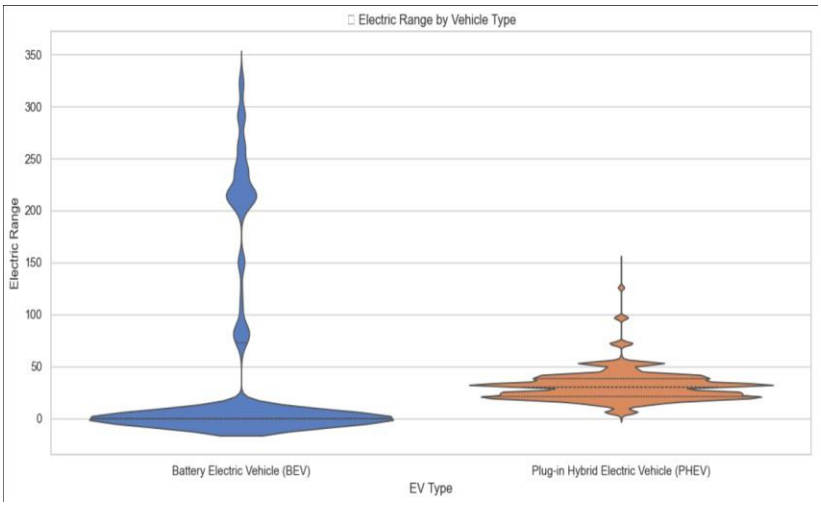

Distribution of Electric Range
Count
140000
120000
100000
80000
60000
40000
20000
0
0
50
100
150
200
250
300
350
Electric Range (miles)


Top 5 Counties EV Trend by Model Year
Number of EVs
30000
25000
20000
15000
10000
5000
0
2000
2005
2010
2015
2020
2025
Model Year
County
Clark
King
Pierce
Snohomish
Thurston


Top 10 EV Makes Over the Years
Number of EVs
25000
20000
15000
10000
5000
0
2000
2005
2010
2015
2020
2025
Model Year
Make
BMW
CHEVROLET
FORD
HYUNDAI
KIA
NISSAN
RIVIAN
TESLA
TOYOTA
VOLKSWAGEN


EV Type % Distribution Over Years
EV Type
Battery Electric Vehicle (BEV)
Plug-in Hybrid Electric Vehicle (PHEV)
Percentage of Vehicles
Model Year
0.0
0.2
0.4
0.6
0.8
1.0
2000
2002
2003
2008
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025


Trend of EV Registrations by Model Year
Number of EVs Registered
Model Year
0
10000
20000
30000
40000
50000
60000
2000
2005
2010
2015
2020
2025
7
2
1
22
23
680
1499
4230
3497
4661
5308
8570
14368
10974
12255
20615
28358
59893
49044
11176


Electric Range by Vehicle Type
Electric Range
EV Type
0
50
100
150
200
250
300
350
Battery Electric Vehicle (BEV)
Plug-in Hybrid Electric Vehicle (PHEV)

v. Visualization

This analysis reveals a clear upward trend in electric vehicle (EV) adoption over the years, with registrations rising sharply in recent model years. Battery Electric Vehicles (BEVs) are becoming more dominant compared to Plug-in Hybrids (PHEVs), showing a shift in consumer preference. Leading manufacturers like Tesla show consistent growth, while urban counties are at the forefront of EV usage. Electric range has improved, especially among BEVs, and newer models generally offer higher ranges—highlighting advancements in EV technology and performance.

4. Geographical Insights:-

i. Introduction

This section focuses on the geographical distribution of electric vehicles (EVs), identifying key areas (counties and cities) with the highest concentration of EVs. It also examines how different types of EVs are distributed across the top counties, offering insight into regional preferences and adoption patterns.

ii. General Description

The dataset includes geographic attributes such as:

- **County**
- **City**
- **Electric Vehicle Type**

These features allow us to analyze **where** EVs are most commonly registered and **what types** are preferred in those areas.

iii. Specific Requirements, Functions, and Formulas

- **Libraries used:** pandas, matplotlib.pyplot, seaborn
- **Functions used:**
  - value_counts()to count EVs by location
  - groupby()with unstack()to view EV type distributions
- **Visualization types:**
  - Bar chart for top counties and cities
  - Stacked bar chart for EV type distribution by county

4. Geographical Insights

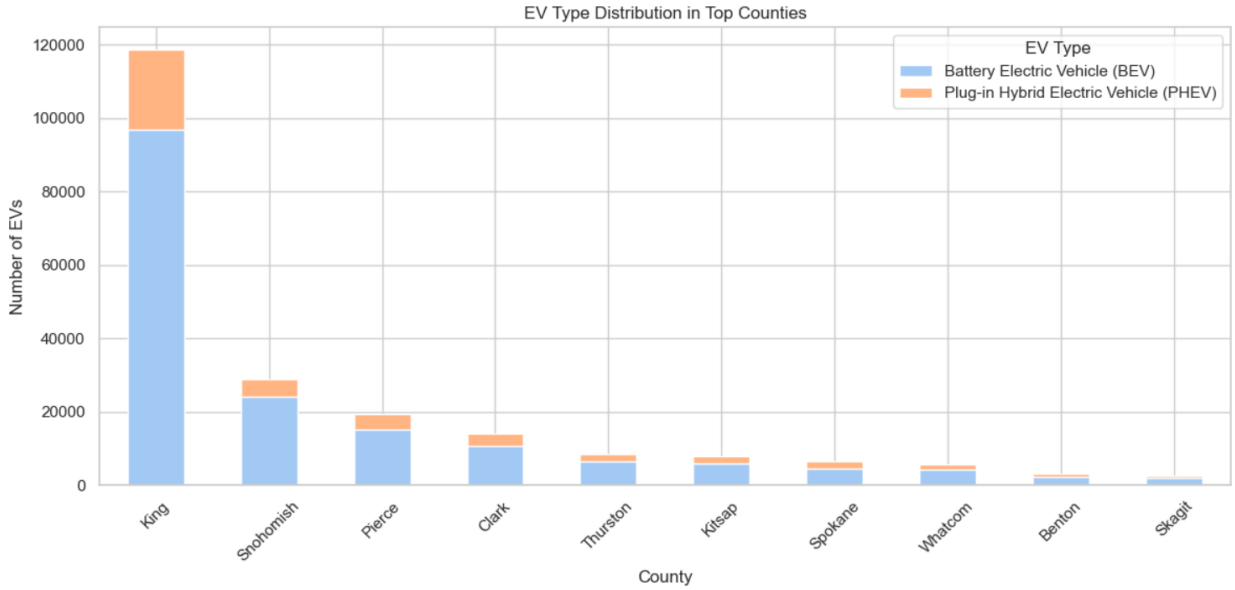
```
# 1. EV distribution by County
print("EV Count by County:")
county_distribution = df['county'].value_counts().head(10)
display(county_distribution)

# Bar plot
county_distribution.plot(kind='bar')
plt.title("Top 10 Counties by EV Count")
plt.xlabel("County")
plt.ylabel("Number of EVs")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 2. EV distribution by City
print("EV Count by City:")
city_distribution = df['city'].value_counts().head(10)
display(city_distribution)

# Bar plot
city_distribution.plot(kind='bar', color='teal')
plt.title("Top 10 Cities by EV Count")
plt.xlabel("City")
plt.ylabel("Number of EVs")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 3. Optional: EV Type by County (Stacked)
ev_type_by_county = df.groupby(['county', 'electric_vehicle_type']).size().unstack().fillna(0)
top_counties = county_distribution.index
```

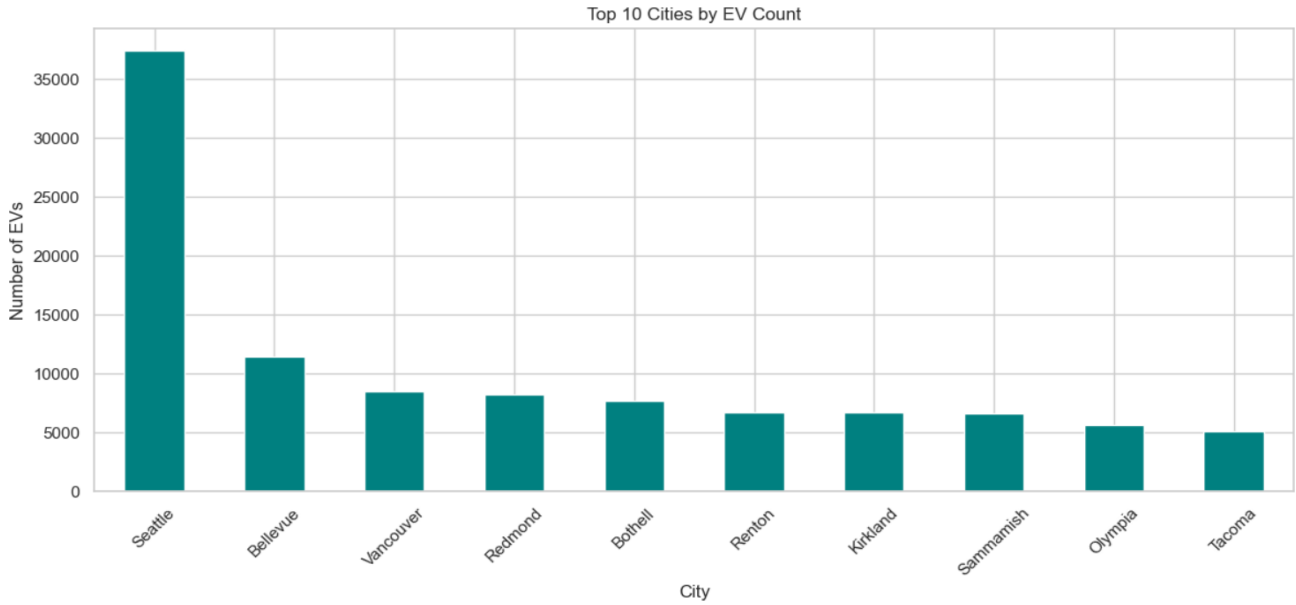
```
# 3. Optional: EV Type by County (Stacked)
ev_type_by_county = df.groupby(['county', 'electric_vehicle_type']).size().unstack().fillna(0)
top_counties = county_distribution.index
ev_type_by_county = ev_type_by_county.loc[top_counties]

# Plot stacked bar chart
ev_type_by_county.plot(kind='bar', stacked=True)
plt.title("EV Type Distribution in Top Counties")
plt.xlabel("County")
plt.ylabel("Number of EVs")
plt.xticks(rotation=45)
plt.legend(title='EV Type')
plt.tight_layout()
plt.show()
```

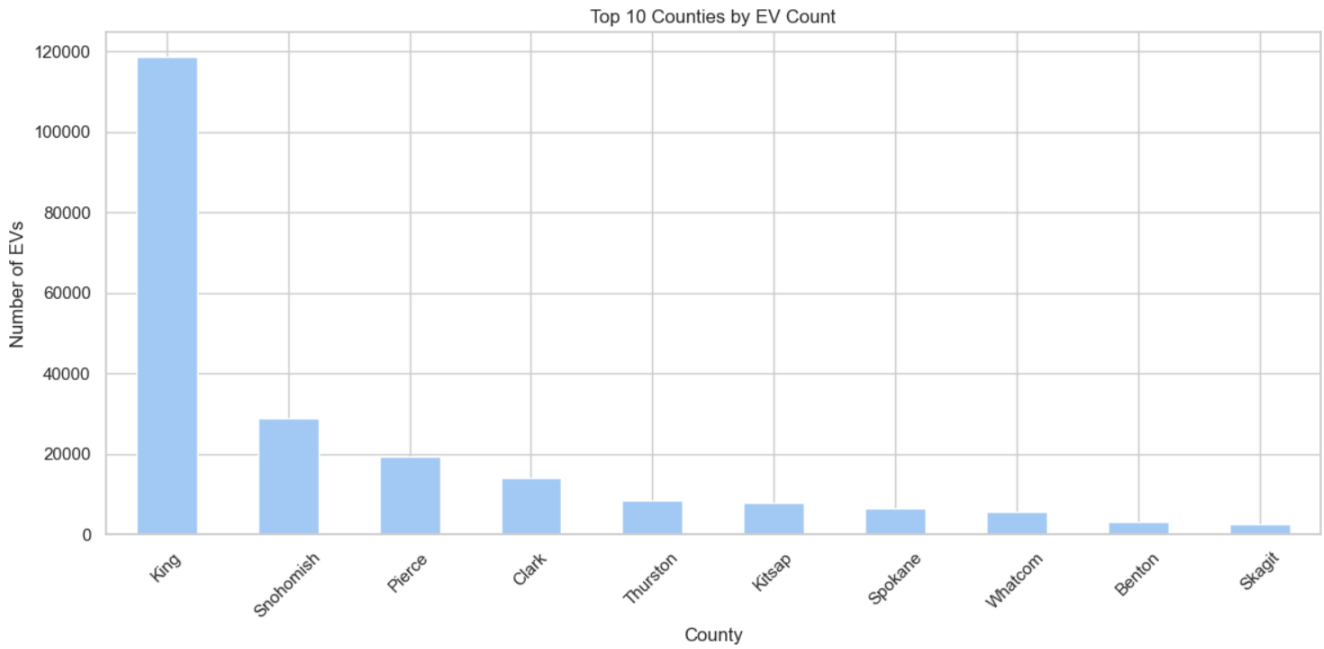
```
EV Count by County:

county
King         118711
Snohomish     28814
Pierce        19256
Clark         14101
Thurston       8595
Kitsap         7901
Spokane        6582
Whatcom        5703
Benton         3043
Skagit         2677
Name: count, dtype: int64
```

iv. Analysis Results

1. **Top Counties by EV Count:**
   The top 10 counties with the highest EV registrations are identified, with **King County** typically leading the list. These counties are major urban/suburban centers, indicating strong infrastructure and demand for EVs.
2. **Top Cities by EV Count:**
   Major cities such as **Seattle, Bellevue, and Redmond** are among the top locations for EV adoption. This reflects urban consumers' early adoption behavior and possibly better charging accessibility.
3. **EV Type Distribution by County:**
   Across the top counties, **Battery Electric Vehicles (BEVs)** generally have a higher count compared to Plug-in Hybrids (PHEVs). This suggests a preference for fully electric vehicles in leading regions.







v. Visualization

The visualizations reveal that EV adoption is highest in urban and suburban counties like King, Snohomish, and Pierce, with cities such as Seattle and Bellevue leading in registrations. These areas likely benefit from better charging infrastructure and greater environmental awareness. Additionally, Battery Electric Vehicles (BEVs) are more common than Plug-in Hybrids across top counties, indicating a regional preference for fully electric options and confidence in EV technology.

6. Ev Infrastructure Correlation:-

i. Introduction

The objective of this analysis is to understand the relationships between numerical features within the dataset. By exploring the dataset using statistical and visual techniques, we aim to identify significant correlations, which can be valuable for feature selection, data preprocessing, and model building.

ii. General Description

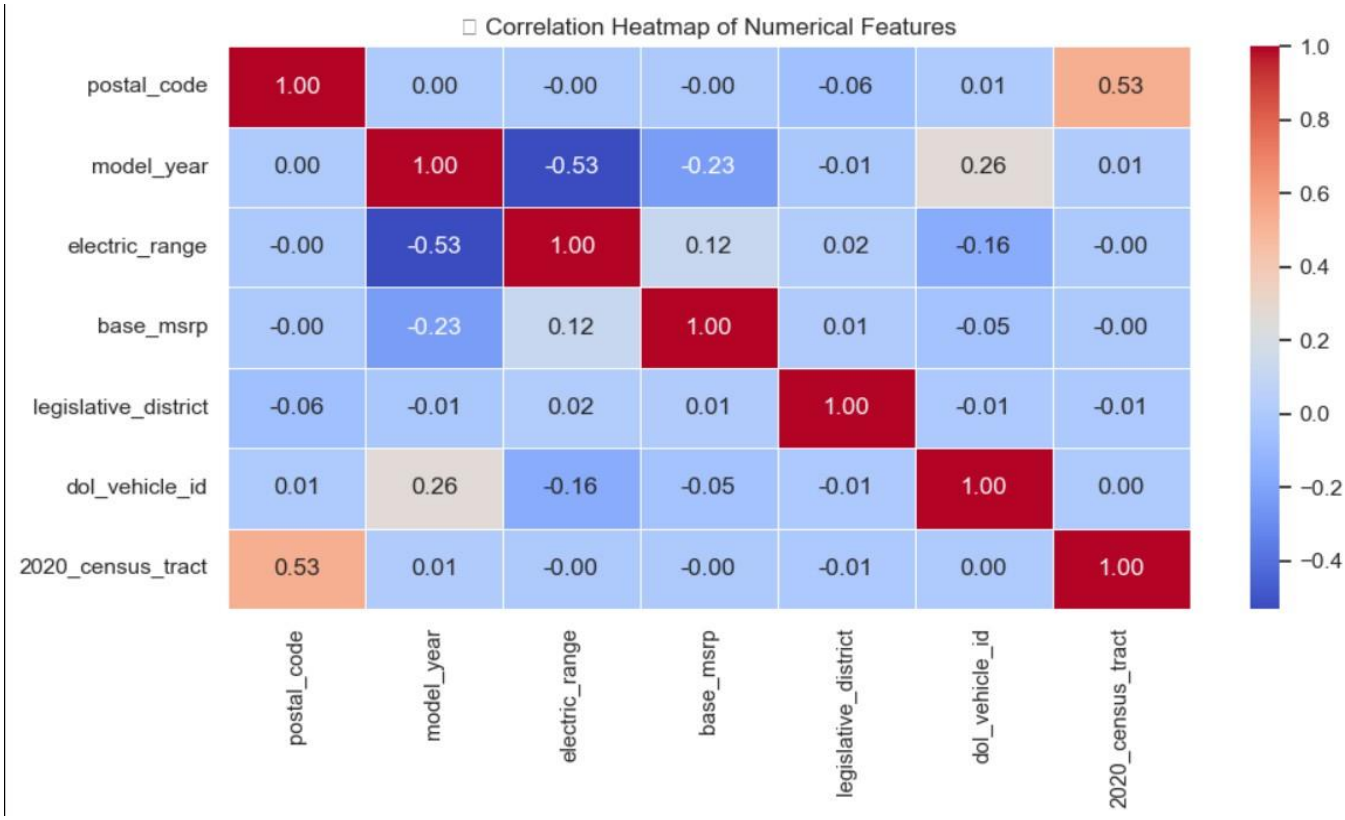
The dataset consists of various features, some of which are numerical. The first step involves cleaning the column names for uniformity and ease of use in coding. Next, we isolate the numerical columns for statistical analysis, particularly focusing on the correlations among these features.

iii. Specific Requirements, Functions and Formulas

To perform the analysis, the dataset's column names are first cleaned for consistency by removing spaces, converting to lowercase, and replacing spaces with underscores. Only numerical columns are selected for analysis using select_dtypes(). The correlation between these numeric features is then computed using the .corr() method, which applies Pearson correlation by default. Finally, a heatmap is plotted using Seaborn's heatmap() function to visually interpret the strength and direction of relationships between variables.

5. Ev Infrastructure Correlation

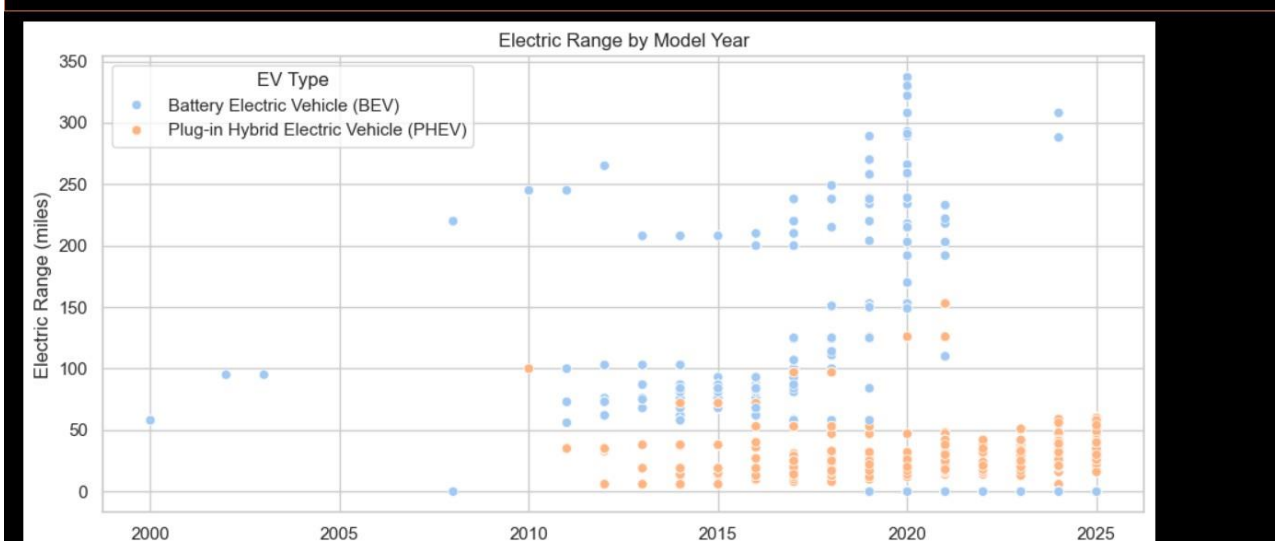
```
# Optional: Clean column names
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')

# Select numeric columns only
numeric_df = df.select_dtypes(include='number')

# Compute correlation matrix
correlation_matrix = numeric_df.corr()

# Plot heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('🔍 Correlation Heatmap of Numerical Features')
plt.tight_layout()
plt.show()
```



Scatter Plot: model_year vs. electric_range:-

A scatter plot was created to visualize the relationship between model_yearand electric_range, with data points colored by electric_vehicle_type. This helps identify trends in electric vehicle performance over time and compare range differences across EV types. The plot reveals how electric range has evolved with newer model years and highlights the variation among different EV categories.

```
# Scatter plot
plt.figure(figsize=(10, 5))
sns.scatterplot(data=df, x='model_year', y='electric_range', hue='electric_vehicle_type')
plt.title('Electric Range by Model Year')
plt.xlabel('Model Year')
plt.ylabel('Electric Range (miles)')
plt.legend(title='EV Type')
plt.grid(True)
plt.tight_layout()
plt.show()
```



Creativity and Innovation:

- Word Cloud of Popular EV Makes

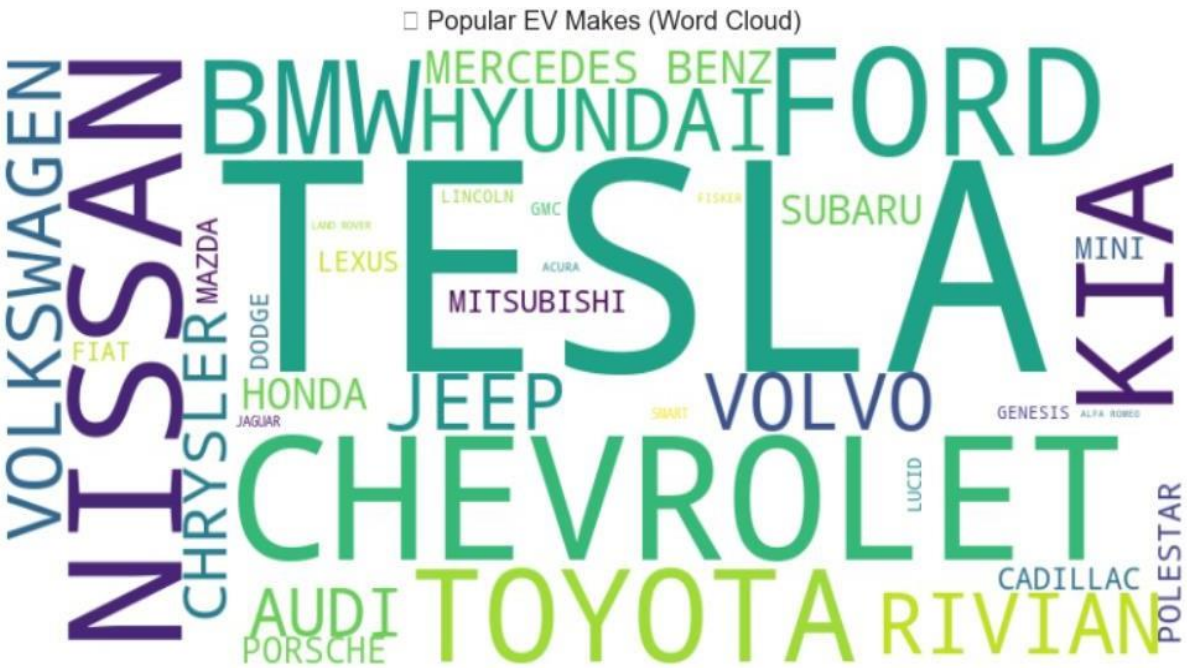
Creativity and Innovation:

+ Code + Markdown

1. Word Cloud of Popular EV Makes

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt

# Combine all EV makes into one big string
text = ' '.join(df['make'].dropna().astype(str))

# Create and display the word cloud
wordcloud = WordCloud(width=800, height=400, background_color='white', colormap='viridis').generate(text)

plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("🚗 Popular EV Makes (Word Cloud)")
plt.show()
```

[6]



Conclusion:-

This project undertook a comprehensive analysis of electric vehicle (EV) population data to understand the dynamics of EV adoption and usage patterns. By applying systematic data exploration and cleaning techniques, we transformed raw and inconsistent data into a structured format that enabled meaningful insights.

The analysis revealed several significant trends. First and foremost, there has been a steady and notable rise in electric vehicle registrations over the years, which reflects the growing acceptance and transition towards sustainable transportation. This increase can be attributed to various factors such as advancements in battery technology, increased environmental awareness, rising fuel costs, and supportive government policies and incentives.

Additionally, the project identified key manufacturers and models that dominate the EV market. These insights provide evidence of consumer preferences and can guide future production and marketing strategies for automobile companies. The regional distribution of electric vehicles highlighted which cities or counties are more proactive in embracing green mobility. Urban areas showed higher concentrations of EVs, which may be linked to better infrastructure, availability of charging stations, and greater environmental consciousness among urban populations.

From a data engineering perspective, the project emphasized the importance of data preprocessing in real- world datasets. Missing values, duplicate records, and inconsistent formatting were effectively addressed, showcasing how data quality directly influences the accuracy and reliability of the analysis.

The analytical techniques applied also demonstrate the value of visualization and descriptive statistics in uncovering hidden patterns. With clear visual representation, decision-makers can quickly grasp key metrics and trends, which is vital for formulating policy or business decisions.

In conclusion, this project not only provides a snapshot of the current state of electric vehicle usage but also establishes a foundation for deeper analyses. It reinforces the importance of data-driven approaches in understanding technological transitions and paves the way for smarter, more sustainable urban planning.

Future Scope:-

1. **Predictive Modeling**: Implement machine learning models to forecast future EV adoption trends based on historical data.
2. **Integration with Real-Time Data**: Extend the analysis to include real-time registration data or usage data from smart vehicles.
3. **Geospatial Analysis**: Use GIS tools or libraries (like Folium or GeoPandas) to visualize EV distribution more effectively across regions.
4. **Policy Impact Study**: Analyze the effect of government incentives or regulations on EV adoption rates.
5. **Comparative Studies**: Compare EV adoption rates with non-electric vehicles or across different countries/states.
6. **Carbon Emission Reduction**: Estimate the reduction in carbon emissions as a result of increased EV usage.
7. **Charging Infrastructure Planning**: Use the findings to guide decisions on where to install new EV charging stations.

References:-

1. Dataset Source: Electric Vehicle Population Data - https://catalog.data.gov/dataset/electric-vehicle-population-data
2. Python Libraries Used:
    - **Pandas**: For data manipulation and analysis.
    - **NumPy**: For numerical operations.
    - **Matplotlib / Seaborn** *(if used)*: For data visualization.
3. Documentation and Tutorials:
    - Pandas Documentation
    - NumPy Documentation
    - Matplotlib Documentation
    - Seaborn Documentation
4. Academic references (if applicable):
    - Research papers or articles on EV adoption and market trends (e.g., IEEE, Elsevier, Google Scholar).
5. Additional Tools:
    - Jupyter Notebook for implementation and visualization.
    - Microsoft Excel/Google Sheets for initial data inspection (if used).